Risk Assessment & Mitigation Plan

Customer Support RAG Chatbot

Project Type: Depi Graduation Project

Domain: AI-Powered E-Commerce Customer Support

Team Size: 5 Members

1 Identified Risks & Mitigation Strategies

The following risks were identified during the project lifecycle. For each risk, a mitigation strategy was designed and implemented to ensure system reliability and response quality.

R1 LLM Hallucinated / Incorrect Answers

AI QUALITY RISK

HIGH IMPACT

RISK DESCRIPTION

The LLM may generate fabricated or incorrect answers that are not grounded in the actual knowledge base, leading to misleading customer support responses and loss of user trust.

MITIGATION STRATEGY

Implemented a **confidence scoring system** with three thresholds (high > 0.5, medium 0.2 - 0.5, low < 0.2). For low-confidence responses, an **answer verification layer** (`verify_answer`) uses the LLM to validate whether the generated answer actually addresses the user's question. Invalid answers are rejected and replaced with a fallback message.

R2 API Abuse & Spam Requests

SECURITY RISK

HIGH IMPACT

RISK DESCRIPTION

Malicious users or bots could flood the API with excessive requests, causing server overload, increased API costs (Groq), and denial of service for legitimate users.

MITIGATION STRATEGY

Integrated **SlowAPI rate limiting** on all critical endpoints: `/chat/ask`, `/chat/voice`, and `/auth/register`. Each endpoint enforces a maximum number of requests per time window. Excessive requests receive a `429 Too Many Requests` response, protecting the system from abuse without affecting normal usage.

R3 Loss of Conversation Context

USER EXPERIENCE RISK

MEDIUM IMPACT

RISK DESCRIPTION

Without conversation memory, the chatbot treats every message as an isolated query. This leads to repetitive responses and failure to understand follow-up questions, degrading the user experience in multi-turn conversations.

MITIGATION STRATEGY

Implemented **conversation history management** that stores and retrieves the **last 6 messages** per conversation from the database. The history is injected into the LLM prompt as context, allowing the model to maintain awareness of previous exchanges and respond coherently across multiple turns within the same conversation.

R4 Unclear & Vague User Queries

RETRIEVAL QUALITY RISK

MEDIUM IMPACT

RISK DESCRIPTION

Users often send vague follow-up messages containing pronouns like *"that"*, *"this"*, *"it"*, or *"like the previous one"*. These queries fail to retrieve relevant documents from the vector database because they lack specific keywords, resulting in poor or irrelevant answers.

MITIGATION STRATEGY

Built a **query rewriting module** (`rewrite_query`) that uses the LLM to transform vague follow-up queries into clear, self-contained questions. The rewriter uses conversation history to resolve pronouns and references, ensuring the rewritten query contains explicit keywords for accurate vector database retrieval.

2 Risk Summary

Overview of all identified risks, their impact levels, and current resolution status.

#	RISK	CATEGORY	IMPACT	MITIGATION	STATUS
R1	LLM Hallucinated Answers	AI Quality	High	Confidence scoring + Answer verification layer	Resolved
R2	API Abuse & Spam	Security	High	SlowAPI rate limiting on critical endpoints	Resolved
R3	Loss of Conversation Context	User Experience	Medium	Chat history (last 6 messages per conversation)	Resolved
R4	Unclear & Vague Queries	Retrieval Quality	Medium	LLM-based query rewriting module	Resolved